

AWS S3 Storage Classes – Complete Interview Cheat Sheet

What is a Storage Class?

An Amazon S3 **Storage Class** defines how your data is stored, its availability, durability, retrieval speed, and storage cost. You choose a storage class based on how frequently the data is accessed and how quickly it must be retrieved.

Storage Class	Access Frequency	Retrieval Time	Multi-AZ	Storage Cost	Retrieval Fee
S3 Standard	Frequent	Milliseconds	Yes	Highest	No
Intelligent-Tiering	Unknown/Changing	Milliseconds	Yes	Varies	No*
Standard-IA	Infrequent	Milliseconds	Yes	Lower	Yes
One Zone-IA	Infrequent	Milliseconds	No	Lower than Std-IA	Yes
Glacier Instant Retrieval	Rare	Milliseconds	Yes	Low	Yes
Glacier Flexible Retrieval	Very Rare	Minutes–Hours	Yes	Very Low	Yes
Glacier Deep Archive	Almost Never	12–48 Hours	Yes	Lowest	Yes

S3 Intelligent-Tiering

Automatically moves objects between access tiers based on **access patterns**, not file type or business use case. If an object becomes active again, AWS automatically moves it back to the appropriate access tier.

Interview Tips

- Standard: Frequent access, no retrieval fee.
- Standard-IA & One Zone-IA: Lower storage cost, retrieval fee applies.
- Glacier classes: Archive storage with retrieval charges.
- One Zone-IA stores data in only one Availability Zone.
- *Intelligent-Tiering has no retrieval fee in Frequent/Infrequent Access tiers; archive tiers incur retrieval charges.